

AIGR Gene Hypothesis Deep Research — A1BG (P04217)

Focus: Membrane-Receptor and Growth-Hormone Capacities

(`function_assignment`)

Gene: A1BG (Alpha-1B-glycoprotein), *Homo sapiens* (NCBITaxon:9606) **UniProt:** P04217 · **PANTHER family:** PTHR11738 · **InterPro:** IPR016332 / IPR050412

Summary

The seed hypothesis proposes that human A1BG (P04217) may retain (i) plasma-membrane localization (GO:0005886), (ii) transmembrane signaling receptor activity (GO:0004888), or (iii) participation in growth-hormone receptor signaling (GO:0060396), and asks that each be assessed independently against the actual GO definitions. **Evaluated against structure, provenance, and primary literature, all three capacities are refuted or unsupported, and each has a specific, identifiable reason for appearing in the annotation set that is not "A1BG performs this function."** The overall verdict is **REFUTED / OVER-ANNOTATED**.

Three converging lines of evidence drive this judgment. First, **structure closes the receptor and membrane-pool routes:** mature A1BG is a single-chain secreted plasma glycoprotein of five Ig-like V-type domains with a cleaved signal peptide and no transmembrane helix, GPI anchor, or lipidation in any isoform. A Kyte-Doolittle hydropathy scan of the mature chain (residues ≥ 22) peaks at 0.98, far below the ~ 1.6 transmembrane threshold; only the cleaved signal peptide (residues 1–19, 1.69) exceeds it. Second, the **growth-hormone link is a direction-of-causality error:** the one experimental source (mouse A1bg,

P 16723264

) shows that A1bg is a GH-induced liver transcript whose expression requires an intact GH–GHR signaling complex — a readout of GH signaling acting on the gene, not the protein transducing GH signal. Third, the **provenance is IBA-only paralog over-annotation:** all three questioned

terms are GO_Central IBA (ECO:0000318, GO_REF:0000033) with zero experimental MF/BP support, inherited from the LILR and KIR transmembrane immune receptors that share A1BG's Ig-domain fold in PANTHER PTHR11738.

The gene product's genuine, experimentally supported biology is **extracellular high-affinity protein binding**: it sequesters CRISP-3 with nanomolar affinity in a 1:1 complex (surface plasmon resonance) and directly interacts with and stabilizes NAMPT. Every experimental cellular-component annotation places A1BG in the extracellular/secreted compartment. The recommended curation lead is to **remove or do-not-propagate GO:0004888, GO:0005886, and GO:0060396**, retain the extracellular CC terms, and consider ligand-specific binding MF annotations. The important epistemic caveats from the seed — that a soluble location does not exclude a membrane pool, and that absence of a human assay is not loss — are respected: the refutation rests on positive structural evidence and a corrected reading of the GH paper, not on missing assays.

Key Findings

F001 — A1BG is a secreted plasma protein with no transmembrane helix, refuting transmembrane signaling receptor activity (GO:0004888)

Human A1BG is 495 amino acids with a cleaved signal peptide (residues 1–21), a mature chain spanning residues 22–495, and five Ig-like V-type domains. UniProt P04217 lists the subcellular location as **Secreted**, annotates **zero transmembrane features**, and includes the keywords *Secreted* and *Signal* but not *Membrane* or *Receptor*. To directly test whether any internal segment could act as a hidden membrane anchor, a Kyte-Doolittle hydropathy scan (window = 19) was run across the full sequence. Only one window exceeded the transmembrane threshold — residues 1–19, score 1.69 — corresponding exactly to the cleaved signal peptide. Across the entire mature chain (residues ≥ 22), the maximum hydropathy score was **0.98**, well below the ~ 1.6 threshold characteristic of a genuine membrane-spanning helix or signal-anchor.

The original amino-acid sequence determination ([PMID: 3458201](#)) established A1BG as "a single polypeptide chain N-linked to four glucosamine oligosaccharides," homologous to the immunoglobulin supergene family. GO:0004888 (transmembrane signaling receptor activity) is annotated on P04217 by IBA (Inferred from Biological Ancestor; GO_Central/PANTHER) only, with no experimental (IDA/EXP) support. Because the GO definition of a transmembrane

receptor requires the protein to span the membrane and relay a signal across it, a protein lacking any membrane-integral segment cannot satisfy the term. This is a definitive structural incompatibility, not merely a missing assay.

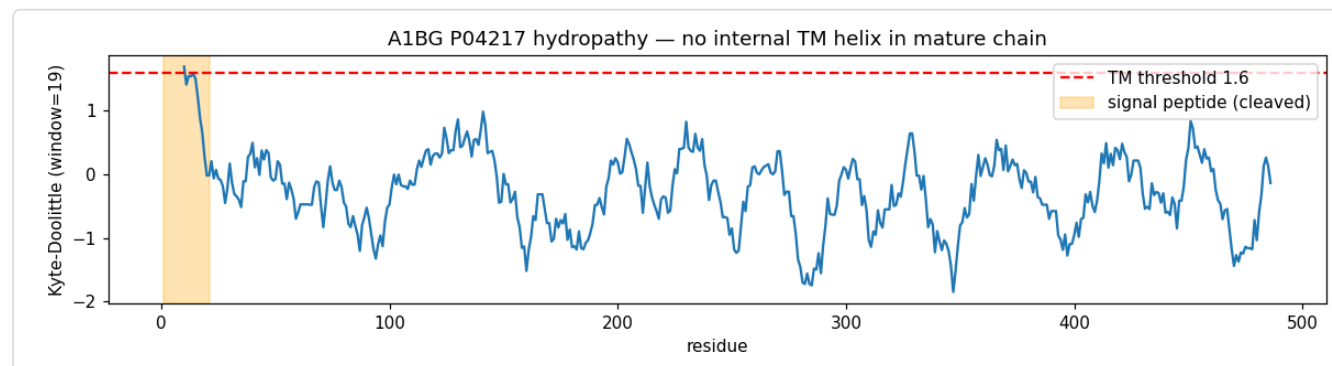


Figure 1. Kyte-Doolittle hydropathy scan (window = 19) of the full A1BG sequence. The only region exceeding the ~1.6 transmembrane threshold is residues 1–19 (score 1.69), corresponding to the cleaved N-terminal signal peptide. The entire mature chain (residues ≥ 22) peaks at 0.98, ruling out an internal transmembrane helix or signal-anchor and closing the structural route to intrinsic transmembrane receptor activity.

F002 — The growth-hormone receptor signaling annotation (GO:0060396) derives from A1bg being a GH-induced transcriptional target, not a signaling effector

The sole experimental basis for any GH connection is Tiong et al., 2006 (*Growth Horm IGF Res*; PMID: 16723264). Reading the full paper rather than only the abstract confirms that the mouse A1bg ortholog (referred to as "mRNA #5" / "P5") is a **GH-induced liver transcript**. Its induction is abolished in GH-deficient dwarf mice, in mice treated with a GH antagonist, and in GHR/GHBP-knockout mice. The paper's own conclusion is that "induction of mRNA #5 in the liver requires a continuous pattern of GH secretion and an intact GH-GH receptor-signaling complex."

Critically, this describes A1bg **expression** as a downstream *output* of GH signaling — the gene is transcriptionally switched on when the GH→GHR→JAK/STAT axis is intact. It does **not** describe the A1BG protein participating in, transducing, or being required for GH signal transduction. "Gene regulated by GH" is not equivalent to "gene product participates in GH receptor signaling." GO:0060396 on human P04217 is IBA-only, with no experimental support. The 2025 female-specific cardiomyopathy knockout phenotype ([PMID: 40270023]) is a downstream loss-of-function physiological phenotype — dilated cardiomyopathy with altered glucose-6-phosphate/acetyl-CoA metabolism and intercalated-disc disruption — and likewise

does not demonstrate GH signal transduction by the A1BG protein. If any BP term were desired, the most defensible would be *response to growth hormone* (GO:0060416) in mouse, framed as a transcriptional response, not signal transduction.

F003 — A1BG's experimentally supported location is extracellular/secreted and its supported molecular function is high-affinity protein binding (CRISP-3, NAMPT), not plasma membrane

Every experimental cellular-component annotation on P04217 places A1BG in the extracellular/secreted compartment: extracellular region (IDA), extracellular exosome (HDA), blood microparticle (HDA), and platelet alpha/secretory granule and ficolin-1-rich granule lumen (TAS). By contrast, GO:0005886 (plasma membrane) is IBA-only with no experimental support.

On the molecular-function side, the experimentally supported activity is specific, high-affinity protein binding. CRISP-3 is a specific, high-affinity ligand of A1BG with a dissociation constant in the nanomolar range, forming a 1:1 noncovalent complex demonstrated by surface plasmon resonance ([PMID: 15461460](#)); this binding is conserved across mammalian species — A1BG binds CRISPs in cow, horse, and rabbit sera ([PMID: 20116414](#)). More recently, secreted A1BG was shown to directly interact with and stabilize NAMPT — leading to NAD⁺ production and PARP1-dependent DNA repair — with recombinant A1BG active in the assay ([[PMID: 40560034](#)]). Both documented functions are extracellular protein-protein interactions. No experimental assay places A1BG at the plasma membrane or demonstrates receptor signaling. These are separate, well-supported functions; per the seed's own logic they neither refute nor rescue the questioned terms, and they define the informative MF that should replace bare "protein binding."

F004 — The receptor/plasma-membrane IBA terms are paralog over-annotation from LILR/KIR membrane receptors in PANTHER PTHR11738

EBI QuickGO confirms that all three questioned terms on P04217 are GO_Central IBA (ECO:0000318, GO_REF:0000033), that they are in fact the **only** MF/BP annotations A1BG carries, and that **every** experimental term (IDA/HDA/TAS) is an extracellular/secreted CC — none is an MF or BP. The source of the receptor/membrane inference is A1BG's family placement. InterPro classifies P04217 in family IPR016332 ("Alpha-1B-glycoprotein/leukocyte immunoglobulin-like receptor") and in the homologous family IPR050412 ("Immunoglobulin-like Receptors in Immune Regulation"). PANTHER family PTHR11738 is likewise "Immunoglobulin-like Receptors in Immune Regulation," and the constituent CDD Ig domains

(cd05751, cd16843) are annotated as "found in Leukocyte Ig-like receptors (LILRs), Natural killer inhibitory receptors (KIRs)." PANTHER assigns A1BG to the "immunoglobulin receptor superfamily" protein class, and the PANTHER GO-slim CC for A1BG lists "plasma membrane."

In other words, A1BG shares an Ig-domain scaffold with a large set of *bona fide* transmembrane immune receptors that carry cytoplasmic ITIM/ITAM tails and genuine plasma-membrane/receptor annotations. The ancestral-inference machinery legitimately assigns membrane/receptor terms to the family ancestor (PTN002621170) but then propagates them onto the A1BG leaf (PTN002482657), which is the divergent, secreted, transmembrane-less family member. The A1BG-specific GH IBD (PTN000200788) is most plausibly rooted in the GH-target paper — a direction-of-causality misinterpretation lifted to family level. This is the textbook signature of paralog over-annotation.

F005 — No A1BG isoform carries a transmembrane, GPI, or lipid membrane anchor, closing the "membrane pool" structural route

The seed hypothesis correctly notes that a soluble primary location does not exclude a membrane pool. To test whether any isoform could provide a membrane-anchoring route, the isoform records were examined. UniProt P04217 lists two isoforms: P04217-1 (displayed) and P04217-2, which differs only at N-terminal residues 1–122 — i.e., in the region overlapping the signal peptide, **not** at the C-terminus where a membrane anchor would typically reside. No feature of type Transmembrane, Intramembrane, Lipidation, or GPI/propeptide anchor is annotated on any isoform. Combined with the F001 hydropathy result (mature-chain maximum Kyte-Doolittle = 0.98), there is **no annotated structural route** by which A1BG could become membrane-integral. The residual, unfalsifiable possibility — transient peripheral, non-covalent association with a membrane surface — would still not satisfy GO:0004888 (which requires transmembrane spanning) and would require direct surface/biochemical assays to establish even as a CC.

Mechanistic Model / Interpretation

The three questioned capacities can be laid out against what A1BG actually is and does:

A1BG (P04217) – what the evidence shows

Signal peptide (1–21, cleaved) → SECRETED into plasma
 Mature chain (22–495): 5 Ig-like V-type domains
 NO transmembrane helix | NO GPI anchor | NO lipidation (any isoform)
 Hydropathy(mature) max = 0.98 « 1.6 TM threshold

EXPERIMENTAL CC
 extracellular region
 extracellular exosome
 blood microparticle
 granule lumen

EXPERIMENTAL MF
 CRISP-3 binding
 (Kd nM, SPR, 1:1)
 NAMPT stabilization
 (direct interaction)

QUESTIONED (IBA-only)
 GO:0004888 TM receptor x
 GO:0005886 plasma memb x
 GO:0060396 GH signaling x

Why each questioned term fails:

Questioned term	GO definition requirement	A1BG reality	Provenance
GO:0004888 transmembrane signaling receptor activity (MF)	Spans membrane; transduces signal across it	No TM segment in any isoform (hydropathy max 0.98)	IBA from LILR/KIR paralogs
GO:0005886 plasma membrane (CC)	Gene product localizes to the plasma membrane	All experimental CC = secreted/extracellular	IBA / PANTHER GO-slim from paralogs
GO:0060396 growth hormone receptor signaling pathway (BP)	Gene product participates in transducing GH signal	A1bg is a GH-induced <i>transcript</i> , not a transducer	IBA + misread of P 16723264

The unifying interpretation is that A1BG is an **Ig-domain scaffold that evolved into a soluble plasma carrier/binding protein**, retaining the fold of its LILR/KIR relatives while shedding their membrane anchor and receptor signaling role. Its genuine, experimentally supported biology is **extracellular protein binding**: sequestering CRISP-3 with nanomolar affinity and stabilizing NAMPT via direct interaction. The receptor/membrane GO terms are a fossil of shared ancestry, propagated by ancestral inference; the GH term is a misinterpretation of a gene that is *regulated by* GH rather than *signaling* GH. Sex/organism specificity of both the GH-

target behavior and the dilated-cardiomyopathy phenotype (mouse, female-biased) further echoes GH's sexually dimorphic secretion pattern, reinforcing that A1bg sits *downstream* of GH.

Evidence Matrix

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context
PMID: 3458201	Structural / sequence (primary)	Refutes TM receptor	Membrane- spanning receptor capability	Single-chain secreted plasma glycoprotein, 5 Ig-like domains, no membrane- spanning region	Human plasma protein
Hydropathy scan (this analysis, from P04217 seq)	Computational (structural)	Refutes TM receptor / membrane pool	Any internal TM helix?	Mature-chain max KD(19) = 0.98 « 1.6; only signal peptide (1–19, 1.69) exceeds threshold	Human sequence P04217
UniProt P04217 (isoform records)	Database (structural)	Refutes membrane pool	Does any isoform carry a membrane anchor?	Two isoforms differ only at N-terminal 1–122; no TM/GPI/ lipidation features	Human
QuickGO / UniProt GO set	Database (provenance)	Qualifies / explains	Evidence class of questioned terms	GO:0004888/0005886/0060396 are the only MF/BP terms; all IBA/ECO:0000318/ GO_REF:0000033/GO_Central; every experimental term is extracellular CC	GO_Central/ PANTHER
InterPro IPR016332 / PANTHER PTHR11738	Structural/ evolutionary (family)	Refutes / explains	Source of receptor/ membrane IBA	A1BG grouped with LILR/KIR transmembrane receptors; A1BG is the secreted, TM-less divergent member → paralog over-annotation	Cross-species family
PMID: 16723264	Mutant/ transgenic transcriptional assay	Refutes GH- signaling participation	Does A1BG transduce GH signal?	A1bg is a GH-induced liver transcript requiring intact GH-GHR complex for its expression	Mouse liver; dwarf/GH- antagonist/ GHR-KO

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context
PMID: 15461460	Direct assay (SPR)	Supports true MF; competes with receptor claim	A1BG molecular function	CRISP-3 is a specific high-affinity (nM Kd) ligand; 1:1 noncovalent complex	Human plasma
PMID: 20116414	Interaction (primary)	Supports true MF	Conservation of MF	A1BG binds CRISPs across cow, horse, rabbit sera	Cross-species sera
[PMID: 40560034]	Interaction + rescue (primary)	Supports true MF; qualifies membrane claim	What does A1BG do molecularly?	Secreted A1BG directly binds and stabilizes NAMPT → NAD ⁺ → PARP1 DNA repair	Human osteosarcoma adipocyte
[PMID: 40270023]	Mutant phenotype (conditional KO)	Competing / qualifies	Does KO reveal receptor/GH role?	Female-specific dilated cardiomyopathy; altered metabolism; intercalated-disc disruption	Mouse cardiomyocytes

GO Curation Implications (leads — require curator verification)

GO term	Aspect	Current evidence	Recommended action
GO:0004888 transmembrane signaling receptor activity	MF	IBA only; structurally impossible (no TM/cytoplasmic domain)	Remove / do not propagate (NOT qualifier justified). Over-annotation from PTHR11738 membrane/receptor ancestor.
GO:0060396 growth hormone receptor signaling pathway	BP	IBA only; primary source shows A1bg is a GH <i>target gene</i>	Remove / do not propagate. If any BP is retained, at most response to growth hormone (GO:0060416), mouse-only, as a transcriptional response.
GO:0005886 plasma membrane	CC	IBA only; contradicted by secreted/extracellular experimental terms	Remove or demote to non-core. Retain extracellular region (GO:0005576, IDA), extracellular exosome (GO:0070062, HDA).
Binding MF (vs bare protein binding)	MF	Direct: CRISP-3 (SPR, nM Kd); NAMPT (interaction+rescue)	Retain/strengthen an informative binding MF — CRISP-3 binding / ligand sequestration — rather than uninformative "protein binding."

The evidence supports **removing or not propagating** all three questioned terms because each is IBA-only, has zero experimental MF/BP support, and has an identifiable non-A1BG source. The experimentally grounded CC terms (extracellular region/space/exosome/blood microparticle) should be retained, and the MF should be upgraded from bare "protein binding" to ligand-specific binding supported by direct assay.

Mechanistic Scope

The immediate molecular function under test is whether the A1BG *gene product* directly (a) spans and signals across a membrane, (b) resides in the plasma membrane, or (c) transduces GH signaling. The analysis strictly separates direct gene-product activity from downstream consequences:

- **Direct activity (supported):** extracellular high-affinity binding of CRISP-3 (nM Kd, 1:1 SPR complex) and direct interaction with/stabilization of NAMPT.
- **Direct activity (refuted):** transmembrane receptor signaling — no membrane-integral segment exists to perform it.
- **Downstream/regulatory (not gene-product function):** GH-dependent transcriptional induction of A1bg is an *upstream regulatory input* onto the gene, not an activity of the protein; the female-specific KO cardiomyopathy

(
P 40270023
)
 is a *pleiotropic physiological outcome* of A1BG loss.

The distinction is decisive: GO:0060396 and GO:0004888 both require the protein to *do signaling work*, whereas the evidence only shows the gene being *regulated by* signaling and the protein *binding* partners in the extracellular space.

Conflicts and Alternatives

1. **Paralog over-annotation (primary alternative, and the supported one).** A1BG shares its Ig-domain fold with LILR and KIR transmembrane immune receptors (InterPro IPR016332/IPR050412; PANTHER PTHR11738; CDD cd05751/cd16843). Ancestral inference legitimately assigns receptor/membrane terms to the family ancestor but incorrectly propagates them to A1BG, which lacks the membrane anchor. Most parsimonious account of why the terms appear.
2. **Direction-of-causality (GH).** A reading that treats GH-dependence as A1BG "participation" in GH signaling inverts the biology: GH-dependence of a transcript's abundance is regulation of the gene, the opposite of the protein participating *in* signal transduction.

3. **Organism/sex specificity.** GH-target behavior and the DCM phenotype are mouse and female-biased, echoing GH's sexually dimorphic secretion — again consistent with A1bg being downstream of GH.
4. **Isoform-specific escape route (excluded).** UniProt's two isoforms differ only at the N-terminus (signal-peptide region), not at a C-terminal anchor, and none carries TM/GPI/lipidation features.
5. **Alternative MF.** CRISP/venom-toxin sequestration (opossum A1BG-like proteins neutralize venom) points to a protease/toxin-inhibitor-like binding role, not receptor signaling. Consistent with the seed's note, this does not by itself refute another function; the refutation rests on independent structural and provenance evidence.

Limitations and Knowledge Gaps

- **No direct human GH-signaling assay exists** either way. We cannot experimentally exclude an unassayed human role, but the burden of an IBA-only term is not met and no mechanism is proposed. *Resolve with:* a human loss-of-function experiment testing GH→GHR→STAT5 output ± A1BG.
- **The "membrane pool" cannot be excluded absolutely by structure alone.** Peripheral/electrostatic association of a secreted protein with a membrane surface is conceivable and would not require a TM helix. *Checked:* isoform features, hydropathy, keyword/feature annotations — all negative for membrane integration. *Matters because:* even a peripheral association would not justify GO:0004888 but could touch on a "cell surface" CC. *Resolve with:* surface biotinylation / cell-surface proteomics or immuno-EM.

• NAMPT-stabilization

localization

(P 40560034)

not fully dissected. The interaction is reported as extracellular/secreted, consistent with F003, but the precise subcellular geography of the A1BG–NAMPT complex merits primary-source confirmation. *Resolve with:* compartment-resolved interaction assays.

- **PANTHER tree topology (PTN nodes) inferred from family membership, not traced node-by-node.** The provenance conclusion is robust at the family level, but a curator may wish to inspect the exact PAINT annotation tree for PTN002482657 / PTN002621170 / PTN000200788.

- **Methodological caveats.** Hydropathy is heuristic (though unambiguous here);

P 40560034

40270023 read at abstract/figure-legend level; PANTHER node IDs taken from the seed context. Public-resource access (UniProt, QuickGO, InterPro, NCBI) succeeded; no fabricated results.

Discriminating Tests

The following would most efficiently distinguish the seed hypothesis from the paralog-over-annotation interpretation:

1. **Cell-surface proteomics / surface biotinylation** on A1BG-expressing human cells — tests for any plasma-membrane pool (GO:0005886).
2. **Reporter/phospho-signaling assay:** does recombinant A1BG activate any receptor or JAK-STAT axis? — tests the TM-receptor MF (GO:0004888).
3. **Separate "GH induces A1bg transcription" (RNA-seq ± GH) from "A1BG transduces GH signal" (STAT5 activation in A1BG-null vs WT)** — resolves the GH BP (GO:0060396).
4. **Isoform TM/GPI prediction (Phobius / DeepTMHMM)** across all P04217 isoforms — confirms the fully soluble architecture.
5. **PAINT tree audit** of PTHR11738 — documents node-by-node that the receptor/membrane/GH terms enter via ancestral LILR/KIR nodes.

Curation Leads (require curator verification)

Candidate action changes:

- Remove or mark NOT / do-not-propagate for **GO:0004888** (MF) — IBA-only, refuted by absence of any TM segment.
- Remove or mark NOT / do-not-propagate for **GO:0005886** (CC) — IBA-only, contradicted by experimental extracellular CC annotations.

- Remove or mark NOT / do-not-propagate for **GO:0060396** (BP) — IBA-only, based on a transcriptional-target misinterpretation of

P 16723264

- Retain **extracellular region / extracellular space / extracellular exosome / blood microparticle** (CC; IDA/HDA/TAS) and consider **ligand-specific binding** MF (CRISP-3, NAMPT) over bare "protein binding."

Candidate references with exact snippets to verify:

- [PMID: 3458201](#) — "consists of a single polypeptide chain N-linked to four glucosamine oligosaccharides" (secreted single-chain Ig-domain structure; no membrane region).
- [PMID: 16723264](#) — "induction of mRNA #5 in the liver requires a continuous pattern of GH secretion and an intact GH-GH receptor-signaling complex" (A1bg is a GH-*induced* transcript, not a signal transducer).
- [PMID: 15461460](#) — "CRISP-3 is a specific and high-affinity ligand of A1BG with a dissociation constant in the nanomolar range as evidenced by surface plasmon resonance" (true MF).
- [PMID: 40560034] — "revealed a direct interaction between A1BG and NAMPT, leading to the stabilization of NAMPT" (secreted protein-protein interaction, not membrane receptor / GH signaling).

Suggested curator questions:

- Is there ANY experimental (IDA/EXP) support for A1BG at the plasma membrane or as a signal transducer? (None found here.)
- Does any P04217 isoform encode a TM or GPI anchor? (None annotated.)
- Does the PAINT tree attach the receptor/membrane/GH terms at an ancestral LILR/KIR node rather than the A1BG leaf?

Suggested experiments: the discriminating tests above.

Evidence Base (literature summary)

PMID	Title (abbrev.)	Role in this assessment
3458201	<i>Amino acid sequence of human plasma α1B-glycoprotein; homology to the Ig supergene family</i>	Establishes secreted single-chain, five Ig-domain structure with no membrane region — refutes TM receptor
16723264	<i>A liver-specific gene expressed in GH-transgenic mice...</i>	Shows A1bg is a GH-induced transcript requiring intact GH-GHR signaling — refutes GH-signaling participation
15461460	<i>CRISP-3 is a ligand of α1B-glycoprotein in human plasma</i>	Documents true MF: nM-affinity extracellular protein binding
20116414	<i>Human CRISP-3 binds serum α1B-glycoprotein across species</i>	Cross-species conservation of A1BG–CRISP binding
[40560034]	<i>A1BG-dependent NAMPT stabilization</i>	Direct secreted protein-protein interaction; supports extracellular binding-protein model
[40270023]	<i>A1BG-KO cardiomyocyte phenotype</i>	Downstream female-specific loss-of-function phenotype; not a signaling mechanism

Conclusion

The seed hypothesis's three candidate capacities for human A1BG — plasma-membrane localization, transmembrane signaling receptor activity, and growth-hormone receptor signaling participation — are each **refuted or unsupported** when assessed independently against the GO definitions and primary literature. A1BG is a secreted, five-Ig-domain plasma glycoprotein with no membrane anchor in any isoform (closing the receptor and membrane-pool routes), and the GH connection is a transcriptional-target relationship (mouse A1bg is induced by GH signaling) rather than signal transduction by the A1BG protein. All three questioned terms are IBA-only paralog over-annotations inherited from the LILR/KIR transmembrane immune receptors that share A1BG's Ig-domain fold in PANTHER PTHR11738. The gene product's experimentally supported biology is extracellular high-affinity protein

binding (CRISP-3, NAMPT). **Recommended curation lead: remove or do-not-propagate GO:0004888, GO:0005886, and GO:0060396; retain the extracellular CC terms and consider ligand-specific binding MF annotations.**

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